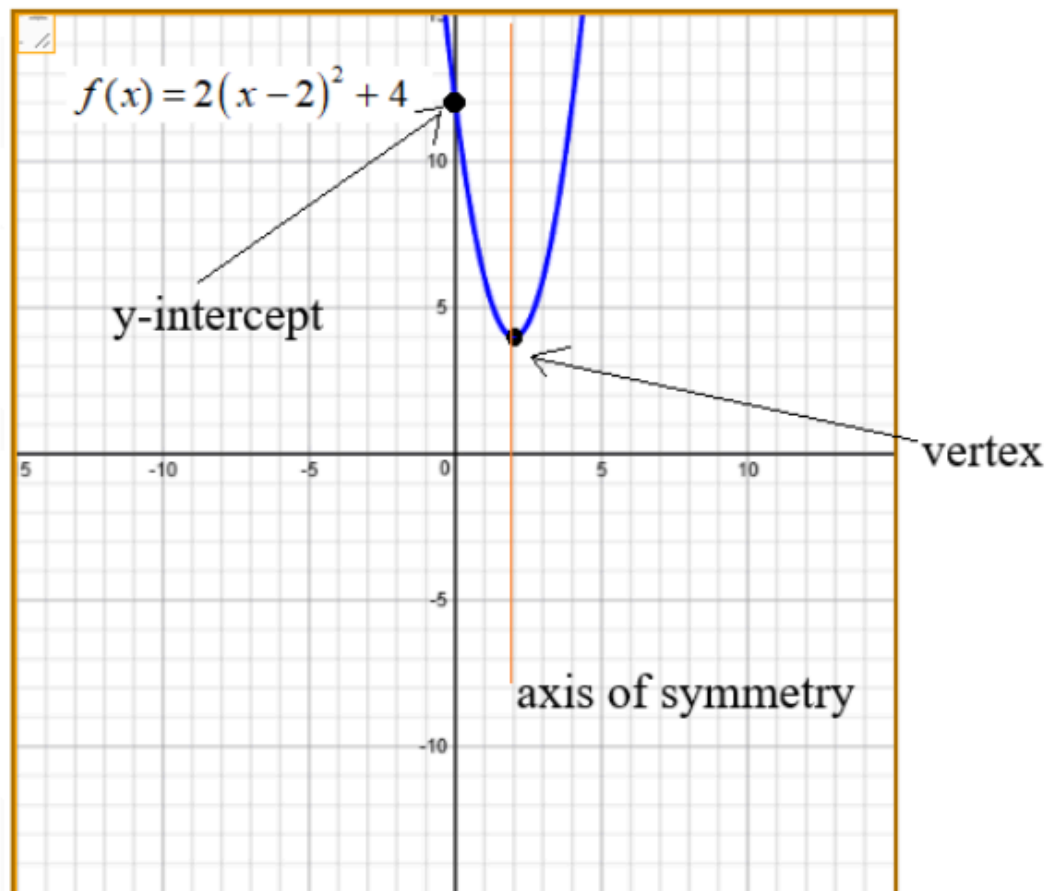


Math 1314 Lesson 20

1) $f(x) = 2(x - 2)^2 + 4$

- a) Find the vertex.
- b) Find x -intercept and y -intercept.
- c) Find the axis of symmetry.
- d) Find the domain and range of the function $f(x)$.
- e) $f(x)$ has a minimum at $x = \underline{\quad ? \quad}$



2) $f(x) = -3(x + 2)^2 - 5$

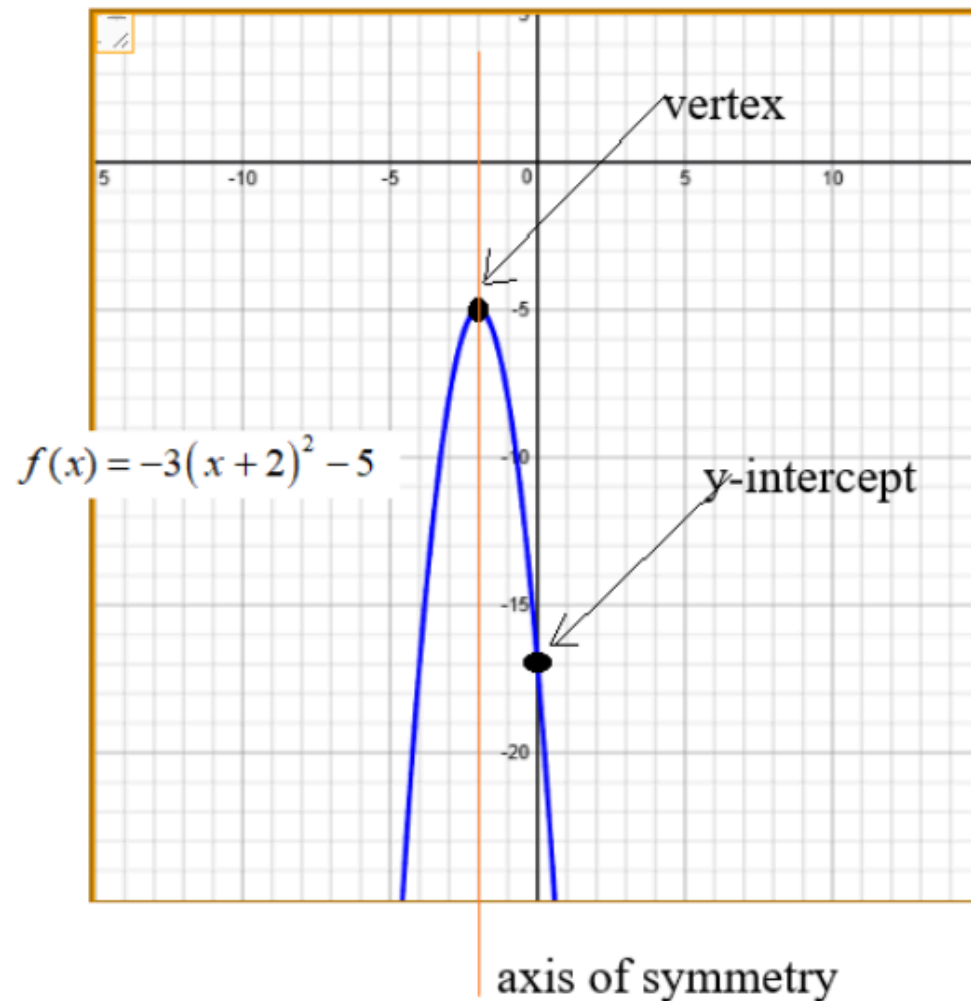
a) Find the vertex.

b) Find x -intercept and y -intercept.

c) Find the axis of symmetry.

d) Find the domain and range of the function $f(x)$.

e) $f(x)$ has a maximum at $x = \underline{\hspace{1cm}}?$



3) $f(x) = 2x^2 - x + 3$

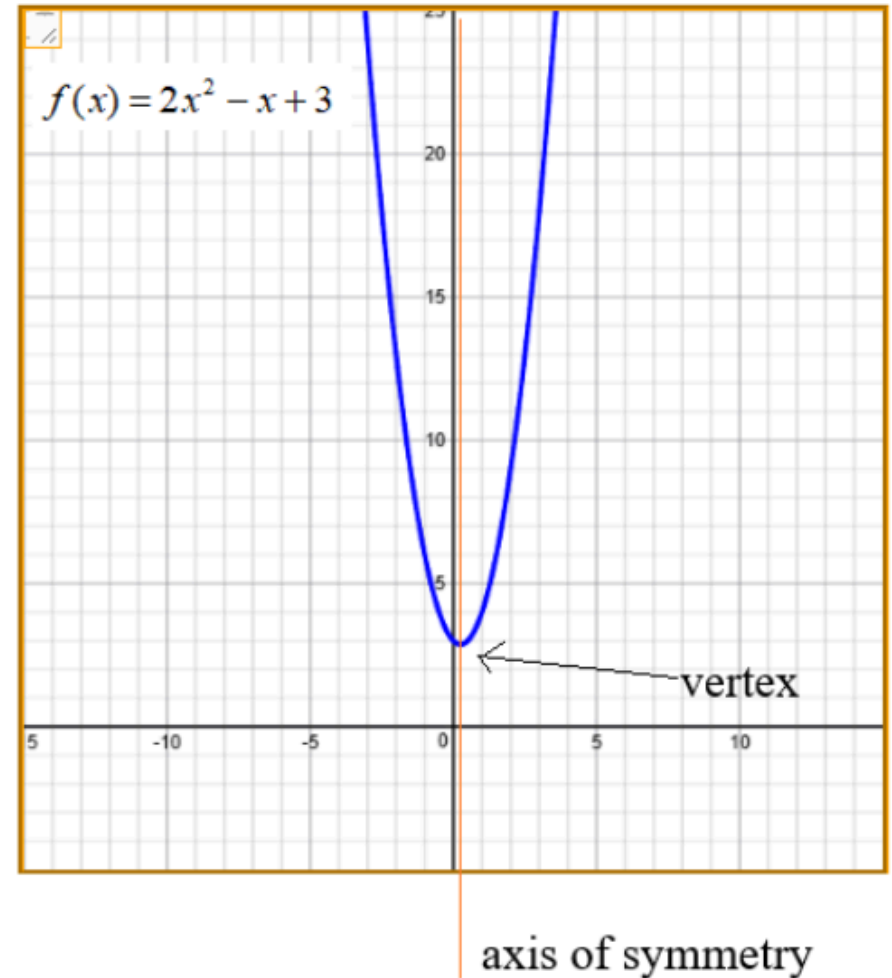
a) Find the vertex.

b) Find x -intercept and y -intercept.

c) Find the axis of symmetry.

d) Find the domain and range of the function $f(x)$.

e) $f(x)$ has a minimum at $x = \underline{\hspace{1cm}} ?$



4) $f(x) = -6x^2 - 11x + 10$

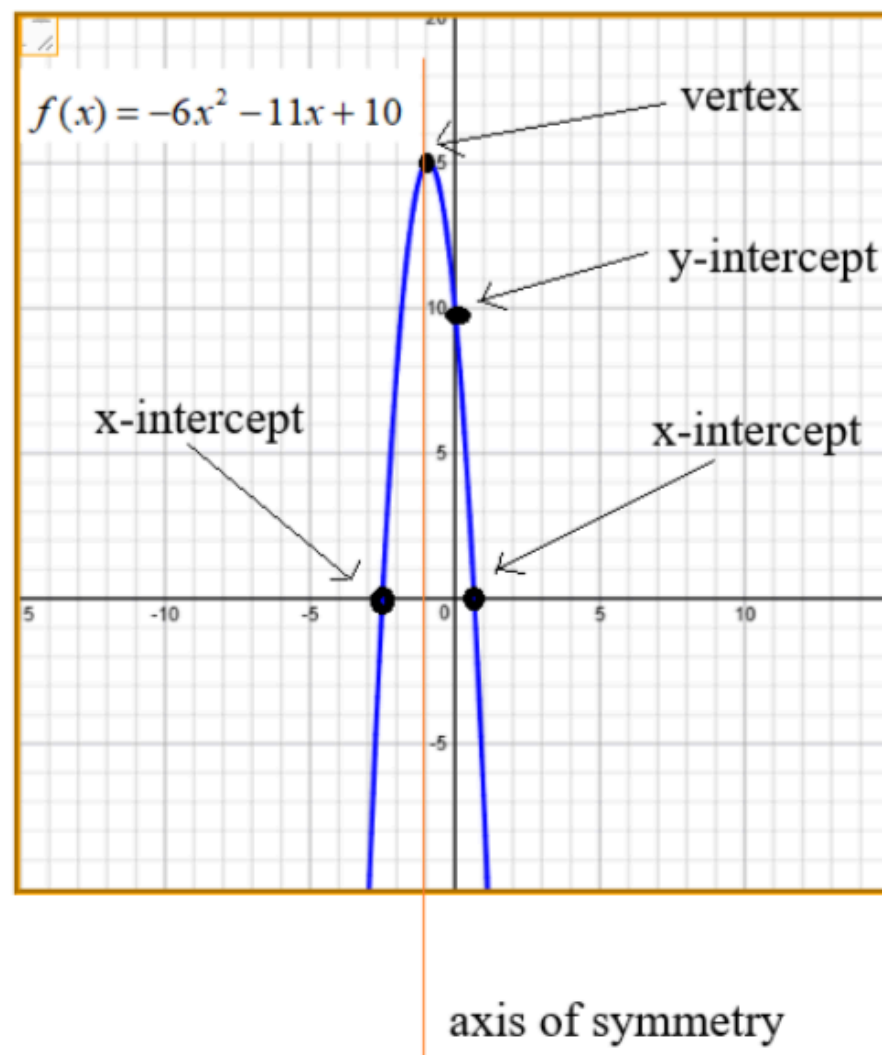
a) Find the vertex.

b) Find x -intercept and y -intercept.

c) Find the axis of symmetry.

d) Find the domain and range of the function $f(x)$.

e) $f(x)$ has a maximum at $x = \underline{\hspace{1cm}} ?$



5) Profit Function

A company manufactures computers and suppose that the profit function is

$P(x) = -3x^2 + 290x - 2500$, where x represents the number of computers sold.

How many computers should be sold to maximize the profit?

- a) Find the profit from selling 5 computers. $P(5) = \underline{\hspace{2cm}}?$
- b) Find the profit from selling 7 computers. $P(7) = \underline{\hspace{2cm}}?$
- c) Find the profit from selling 10 computers. $P(10) = \underline{\hspace{2cm}}?$
- d) Find the profit from selling 50 computers. $P(50) = \underline{\hspace{2cm}}?$
- e) How many computers should be sold to maximize the profit?

Hint: For the quadratic function $f(x) = ax^2 + bx + c$: Vertex is $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

How to graph on TI-84:

Input $Y_1 = 4x^2 - x + 5$

Window Setting: Xmin = -10; Xmax = 100; Ymin = -2700; Ymax = 5000

